

## Subquery 1 — Students scoring above the Average

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (SELECT AVG(Marks) FROM Student);
```

Result Grid

Filter Rows:

Edit:

Export/Import:

|   | Student_ID | Name    | Dept_ID | Marks | City      |
|---|------------|---------|---------|-------|-----------|
| ▶ | 102        | Neha    | 20      | 92    | Mumbai    |
|   | 104        | Manya   | 30      | 88    | Bangalore |
|   | 105        | Tanisha | 20      | 95    | Delhi     |
| ✱ | NULL       | NULL    | NULL    | NULL  | NULL      |

Student 1

×

Output

Action Output

▼

|   | # | Time     | Action                                                                          |
|---|---|----------|---------------------------------------------------------------------------------|
| ✓ | 6 | 20:02:33 | CREATE TABLE Student ( Student_ID INT PRIMARY KEY, Name VARCHAR(50), ...        |
| ✓ | 7 | 20:02:33 | INSERT INTO Department (Dept_ID, Dept_Name) VALUES (10, 'CSE'), (20, 'ECE'),... |
| ✓ | 8 | 20:02:33 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...  |
| ✓ | 9 | 20:02:33 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...      |

## Outer:

CREATE DATABASE IF NOT EXISTS college\_dbms;

USE college\_dbms;

DROP TABLE IF EXISTS Student;

DROP TABLE IF EXISTS Department;

CREATE TABLE Department (

Dept\_ID INT PRIMARY KEY,

Dept\_Name VARCHAR(50)

);

CREATE TABLE Student (

Student\_ID INT PRIMARY KEY,

Name VARCHAR(50),

Dept\_ID INT,

Marks INT,

City VARCHAR(50)

);

INSERT INTO Department (Dept\_ID, Dept\_Name) VALUES

(10, 'CSE'),

(20, 'ECE'),

(30, 'MECH');

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Name, Marks FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);
```

| Result Grid |         |       | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------|-------|--------------|---------|--------------------|
|             | Name    | Marks |              |         |                    |
| ▶           | Neha    | 92    |              |         |                    |
|             | Manya   | 88    |              |         |                    |
|             | Tanisha | 95    |              |         |                    |

| Student 2 x   |          |                                                                                |  |
|---------------|----------|--------------------------------------------------------------------------------|--|
| Output        |          |                                                                                |  |
| Action Output |          |                                                                                |  |
| #             | Time     | Action                                                                         |  |
| ✓ 17          | 20:04:58 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |  |
| ✓ 18          | 20:04:58 | SELECT Name, Marks FROM Student WHERE Marks > (SELECT AVG(Marks) FR...         |  |

### Inner:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
```

City VARCHAR(50)

);

INSERT INTO Department (Dept\_ID, Dept\_Name) VALUES

(10, 'CSE'),

(20, 'ECE'),

(30, 'MECH');

INSERT INTO Student (Student\_ID, Name, Dept\_ID, Marks, City) VALUES

(101, 'Navya', 10, 85, 'Delhi'),

(102, 'Neha', 20, 92, 'Mumbai'),

(103, 'Spoorthi', 10, 76, 'Delhi'),

(104, 'Manya', 30, 88, 'Bangalore'),

(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Name, Marks FROM Student

WHERE Marks >

(SELECT AVG(Marks) FROM Student);

The screenshot displays a database management tool interface. At the top, there is a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar, a table with columns 'Name' and 'Marks' is shown, containing data for Neha (92), Manya (88), and Tanisha (95). The 'Manya' row is highlighted. Below the table, there is a tab labeled 'Student 3'. Underneath the tab, there is an 'Output' section with a dropdown menu set to 'Action Output'. The output log shows two successful actions: an INSERT statement (Action #26) and a SELECT statement (Action #27).

|   | Name    | Marks |
|---|---------|-------|
| ▶ | Neha    | 92    |
|   | Manya   | 88    |
|   | Tanisha | 95    |

Student 3 ×

Output

Action Output

|   | #  | Time     | Action                                                                         |
|---|----|----------|--------------------------------------------------------------------------------|
| ✓ | 26 | 20:06:10 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |
| ✓ | 27 | 20:06:10 | SELECT Name, Marks FROM Student WHERE Marks > (SELECT AVG(Marks) FR...         |

## Subquery 2 — Student(s) who scored the highest marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks = (SELECT MAX(Marks) FROM Student);
```

| Result Grid |            | Filter Rows: | Edit:    |       |       | Export/Import:  |  |
|-------------|------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|---------------------------------------------------------------------------------------------------|--|
|             | Student_ID | Name         | Dept_ID                                                                                                                                                                                                                                                     | Marks | City  |                                                                                                   |  |
| ▶           | 105        | Tanisha      | 20                                                                                                                                                                                                                                                          | 95    | Delhi |                                                                                                   |  |
| *           | NULL       | NULL         | NULL                                                                                                                                                                                                                                                        | NULL  | NULL  |                                                                                                   |  |

Student 4 x

Output

Action Output

| #    | Time     | Action                                                                         |
|------|----------|--------------------------------------------------------------------------------|
| ✓ 35 | 20:08:57 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |
| ✓ 36 | 20:08:57 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...     |

## Inner:

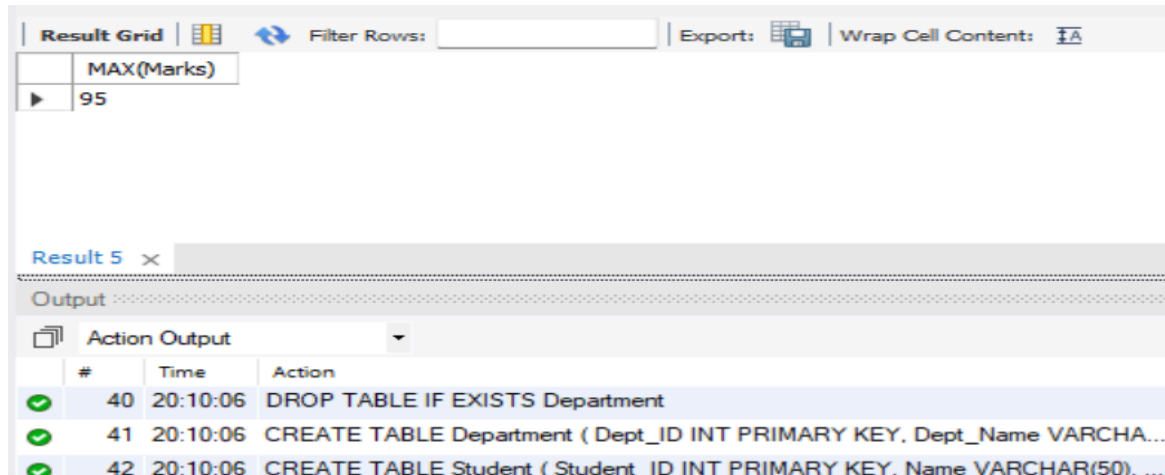
```

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),

```

(104, 'Manya', 30, 88, 'Bangalore'),  
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT MAX(Marks) FROM Student



The screenshot shows a database management interface. At the top, there's a toolbar with 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below this, a table displays the result of a query: a single row with the column 'MAX(Marks)' and the value '95'. Below the result grid, there's a section labeled 'Result 5' with a tab icon. Underneath, there's an 'Output' section with a dropdown menu set to 'Action Output'. This section contains a table with three columns: '#', 'Time', and 'Action'. It lists three actions: dropping the 'Department' table, creating the 'Department' table with columns 'Dept\_ID' (primary key) and 'Dept\_Name' (VARCHAR), and creating the 'Student' table with columns 'Student\_ID' (primary key), 'Name' (VARCHAR(50)), 'Dept\_ID' (INT), 'Marks' (INT), and 'City' (VARCHAR(50)).

|   | MAX(Marks) |
|---|------------|
| ▶ | 95         |

| #  | Time     | Action                                                                                                         |
|----|----------|----------------------------------------------------------------------------------------------------------------|
| 40 | 20:10:06 | DROP TABLE IF EXISTS Department                                                                                |
| 41 | 20:10:06 | CREATE TABLE Department ( Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(50))                                      |
| 42 | 20:10:06 | CREATE TABLE Student ( Student_ID INT PRIMARY KEY, Name VARCHAR(50), Dept_ID INT, Marks INT, City VARCHAR(50)) |

## Subquery 3 — Students scoring above Rahul

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
```

```

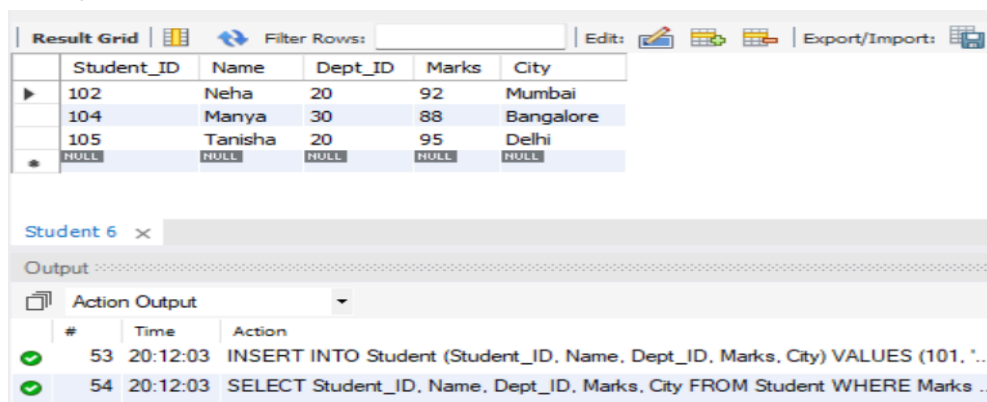
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

```

```

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT Marks FROM Student WHERE Name = 'Rahul' OR Name =
'Navya');

```



The screenshot shows a database management interface. At the top, there's a 'Result Grid' with a table containing 5 columns: Student\_ID, Name, Dept\_ID, Marks, and City. The table has 5 rows of data. Below the result grid, there's an 'Output' section with a tab labeled 'Action Output'. This section shows a log of database actions, including an INSERT statement and a SELECT statement, both marked with green checkmarks indicating successful execution.

| Student_ID | Name    | Dept_ID | Marks | City      |
|------------|---------|---------|-------|-----------|
| 102        | Neha    | 20      | 92    | Mumbai    |
| 104        | Manya   | 30      | 88    | Bangalore |
| 105        | Tanisha | 20      | 95    | Delhi     |
| NULL       | NULL    | NULL    | NULL  | NULL      |

| #  | Time     | Action                                                                         |
|----|----------|--------------------------------------------------------------------------------|
| 53 | 20:12:03 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |
| 54 | 20:12:03 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...     |

## Subquery 4 — Students belonging to the same department as Rahul

```

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)

```

```

);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Dept_ID =
(SELECT Dept_ID FROM Student WHERE Name = 'Navya');

```

| Result Grid                                                                                                                                                                                                                                                          |            |          |         |       |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|---------|-------|-------|
| Filter Rows: <input type="text"/>                                                                                                                                                                                                                                    |            |          |         |       |       |
| Edit:    |            |          |         |       |       |
| Export/Import:                                                                             |            |          |         |       |       |
|                                                                                                                                                                                                                                                                      | Student_ID | Name     | Dept_ID | Marks | City  |
| ▶                                                                                                                                                                                                                                                                    | 101        | Navya    | 10      | 85    | Delhi |
|                                                                                                                                                                                                                                                                      | 103        | Spoorthi | 10      | 76    | Delhi |
| •                                                                                                                                                                                                                                                                    | NULL       | NULL     | NULL    | NULL  | NULL  |

| Student 7     |          | ×                                                                              |
|---------------|----------|--------------------------------------------------------------------------------|
| Output        |          |                                                                                |
| Action Output |          |                                                                                |
| #             | Time     | Action                                                                         |
| 62            | 20:13:12 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |
| 63            | 20:13:12 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Dept_ID...    |

## Subquery 5 — Students who scored above the average CSE marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student WHERE Dept_ID = 10);
```

Result Grid

Filter Rows:

Edit:

Export/Import:

|   | Student_ID | Name    | Dept_ID | Marks | City      |
|---|------------|---------|---------|-------|-----------|
| ▶ | 101        | Navya   | 10      | 85    | Delhi     |
|   | 102        | Neha    | 20      | 92    | Mumbai    |
|   | 104        | Manya   | 30      | 88    | Bangalore |
|   | 105        | Tanisha | 20      | 95    | Delhi     |
| * | NULL       | NULL    | NULL    | NULL  | NULL      |

Student 8

×

Output

Action Output

▼

| #    | Time     | Action                                                                       |
|------|----------|------------------------------------------------------------------------------|
| ✓ 71 | 20:14:18 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '. |
| ✓ 72 | 20:14:18 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks       |

## Subquery 6 — Students who belong to a department having a student from Mumbai

```
CREATE DATABASE IF NOT EXISTS college_dbms;
```

```
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
```

```
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
```

```
Dept_ID INT PRIMARY KEY,
```

```
Dept_Name VARCHAR(50)
```

```
);
```

```
CREATE TABLE Student (
```

```
Student_ID INT PRIMARY KEY,
```

```
Name VARCHAR(50),
```

```
Dept_ID INT,
```

```
Marks INT,
```

```
City VARCHAR(50)
```

```
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
```

```
(10, 'CSE'),
```

```
(20, 'ECE'),
```

```
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
```

```
(101, 'Navya', 10, 85, 'Delhi'),
```

```
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Dept_ID IN
(SELECT Dept_ID FROM Student WHERE City = 'Mumbai');
```

| Result Grid    |            |         |         |       |        |
|----------------|------------|---------|---------|-------|--------|
| Filter Rows:   |            |         |         |       |        |
| Edit:          |            |         |         |       |        |
| Export/Import: |            |         |         |       |        |
|                | Student_ID | Name    | Dept_ID | Marks | City   |
| ▶              | 102        | Neha    | 20      | 92    | Mumbai |
|                | 105        | Tanisha | 20      | 95    | Delhi  |
| ✱              | NULL       | NULL    | NULL    | NULL  | NULL   |

Student 9 ×

Output

Action Output

|   | #  | Time     | Action                                                                         |
|---|----|----------|--------------------------------------------------------------------------------|
| ✓ | 80 | 20:15:54 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |
| ✓ | 81 | 20:15:54 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Dept_ID...    |

## Subquery 7 — Students who scored higher than every ECE student

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
```

```

Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

```

```

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (
SELECT MAX(Marks)
FROM Student
WHERE Dept_ID = 20
);

```

| Result Grid          |            |      |         |       |      |
|----------------------|------------|------|---------|-------|------|
| Filter Rows:         |            |      |         |       |      |
| Edit: Export/Import: |            |      |         |       |      |
|                      | Student_ID | Name | Dept_ID | Marks | City |
| *                    | NULL       | NULL | NULL    | NULL  | NULL |

| Student 10 x  |    |          |                                                                                |  |  |
|---------------|----|----------|--------------------------------------------------------------------------------|--|--|
| Output        |    |          |                                                                                |  |  |
| Action Output |    |          |                                                                                |  |  |
|               | #  | Time     | Action                                                                         |  |  |
| ✓             | 89 | 20:16:50 | INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '... |  |  |
| ✓             | 90 | 20:16:50 | SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...     |  |  |

